

Quantum Rhythm: When Information Breathes in Time

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Abstract

Quantum information is sustained by phase coherence, yet coherence degrades through decoherence and slow detunings. This article proposes an operational hypothesis: a local tempo field, $\Phi(x)$, enters the internal phase evolution of clocks and qubits, while a separate non-negative rate $\Gamma(x)$ captures loss. The admissible imprint of Φ is strictly differential: it appears in ratios, gradients, and residuals after standard GR and known systematics are modeled. In this frame, decoherence reads as desynchronization (loss of phase-lock) to local tempo, yielding testable constraints on Φ through clock networks and T_2 measurements. The goal is not metaphysics but a metrological interface: what to measure, what to subtract, and what would falsify the hypothesis.

Keywords: Chronon Field, quantum coherence, decoherence rate, rhythm, effective clock time, temporal thermodynamics, quantum error correction

Quantum Rhythm — When Information Breathes in Time

I — The fragile rhythm of quantum information

A qubit is a phase oscillator. Its coherence hinges on phase alignment so tight that a tiny fractional detuning of its transition frequency,

$$\delta\nu/\nu \sim 10^{-15}\text{--}10^{-12} \quad (\text{platform \& interrogation time dependent}),$$

measurably shortens T_2 . Single stray photons, millikelvin drifts in a cryostat, or slow magnetic wander can push the phase outside the capture range and erase the state.

Two facts matter operationally:

- Coherence is *temporal alignment*: maintaining a stable phase advance per unit effective clock time.
- Standard formalisms treat time as a silent parameter; they rarely model the *substrate* that sustains phase advance itself.

We adopt a minimal extension: treat time as a local *tempo*. In what follows, the Chronon Field $\Phi(x)$ provides that tempo; loss is carried by a separate rate $\Gamma(x)$. This explicit split (phase vs. loss) lets us write predictions for T_2 and design falsifiable tests across platforms.

II — The hidden tempo beneath coherence

Enter the Chronon Field $\Phi(x)$: not an energetic medium, not a new force, but a local *tempo*. At each spacetime point, $\Phi(x)$ sets the rate at which phase can advance per unit effective clock time. Coherence then means: the qubit's internal oscillation stays entrained to the local tempo.

Operational stance (minimal and falsifiable).

- *Type.* Φ is a scalar with dimension $[\Phi] = \text{s}^{-1}$; geometry and null cones are unchanged, no new $T_{\mu\nu}$.
- *Role.* Φ reparametrizes durations and standards (phase), while loss is carried by a separate non-negative rate $\Gamma(x)$.

- *Testability.* Small spatiotemporal modulations of Φ act as tiny detunings; differential clock comparisons and T_2 measurements can read them once standard GR (gravitational+kinematic) effects are subtracted.

Minimal dictionary (used throughout).

$$d\tau = \Phi^{-1}(x) dt_{\text{lab}}, \quad \underbrace{\dot{\varphi}}_{\text{phase advance}} \equiv \frac{d\varphi}{d\tau}, \quad T_2(x) = \Gamma^{-1}(x).$$

The first line defines an effective clock-time parameter τ relative to a conventional laboratory readout time t_{lab} ; it is a bookkeeping device for internal evolution, not a replacement for GR metric proper time. The second separates *phase* (carried by Φ) from *loss* (carried by Γ). If Φ drifts or varies spatially, phase alignment can fail; coherence then decays even when standard noise is held fixed.

Capture condition (entrainment).

$$|\Delta\Phi| \lesssim \Delta\Phi_c \quad \text{or} \quad |\Delta\Phi/\Phi| \lesssim (\Delta\Phi/\Phi)_c \quad \Rightarrow \quad \text{coherence sustained.} \quad (1)$$

Outside this bandwidth, the qubit falls out of step with the local tempo: information is not destroyed, it *desynchronizes*.

Interlude — Covariant notation & dimensions

Type & units. Φ is a scalar field with $[\Phi] = \text{s}^{-1}$; $\Gamma(x) \geq 0$ is a local loss rate with $[\Gamma] = \text{s}^{-1}$. Geometry and null cones are unchanged; no new $T_{\mu\nu}$.

Worldline form (phase vs. loss). For a trajectory $\gamma : \tau \mapsto x(\tau)$ parametrized by effective clock time τ ,

$$C(\tau) = C_0 \exp\left(i \int_{\gamma} \Phi(x(\tau')) d\tau' - \int_{\gamma} \Gamma(x(\tau')) d\tau'\right), \quad (2)$$

so that the instantaneous relations read

$$\dot{\varphi} = \Phi(x), \quad T_2(x) = \Gamma^{-1}(x).$$

Master operational relations.

$$\boxed{d\tau = \Phi^{-1}(x) dt_{\text{lab}}}, \quad \boxed{J^\mu = \Phi(x) u^\mu}, \quad \boxed{\partial_\mu J^\mu = \Gamma(x)}. \quad (3)$$

Interpretation: (i) Φ reparametrizes durations (clock rates); (ii) J^μ is a *current of coherence* transported by the local four-velocity u^μ ; (iii) Γ closes a continuity law for coherence (source/sink), without adding stress-energy nor altering geodesics.

III — Information as a rhythmic state

Information persists while phase stays *entrained* to the local tempo. A qubit is not a static bit; it is a resonator whose phase advance per unit effective clock time must track $\Phi(x)$.

Entrainment law (minimal). Let $\varphi(\tau)$ be the qubit phase along $\gamma : \tau \mapsto x(\tau)$. We write

$$\dot{\varphi}(\tau) = \Phi(x(\tau)) + \delta_{\text{env}}(\tau), \quad (4)$$

where δ_{env} collects fast, zero-mean perturbations (control, thermal, EM). When $\langle \delta_{\text{env}} \rangle = 0$ and Φ is smooth, the coherence amplitude follows (2) with a loss rate that, at first order, reads

$$\Gamma(x) \simeq \Gamma_{\text{env}}(x) + \xi \Phi(x) + b |\nabla \ln \Phi(x)| \Phi(x), \quad (5)$$

i.e. a platform noise floor plus a tempo term (scalar) and a gradient term (spatial mismatch), as specified in Appendix A.

Rhythmic reading of “information”.

- *Persistence* $\iff$ *phase-lock*: $|\Delta\Phi| \lesssim \Delta\Phi_c$ (or $|\Delta\Phi/\Phi| \lesssim (\Delta\Phi/\Phi)_c$) keeps $|C|$ high.
- *Loss of information* $\iff$ *desynchronization*: Γ increases when Φ drifts or exhibits a gradient that control does not compensate.
- *QEC as re-synchronization*: error correction cycles reduce Γ_{env} (noise) and actively restore alignment to Φ (phase tracking); they cannot mimic a genuine scalar drift of Φ but can *track* it if timing references include differential clocks.

Operational invariant. Define the *rhythmic fidelity* over a window $\Delta\tau$:

$$\mathcal{F}_{\Delta\tau} \equiv \left| \frac{1}{\Delta\tau} \int_{\tau}^{\tau+\Delta\tau} \exp\left(i \int_{\tau}^{\tau'} [\dot{\phi}(s) - \Phi(x(s))] ds\right) d\tau' \right|. \quad (6)$$

Then $\mathcal{F}_{\Delta\tau} \approx 1$ in entrainment, while $\mathcal{F}_{\Delta\tau} \downarrow$ as the mismatch integrates; empirically, $\mathcal{F}_{\Delta\tau}$ tracks T_2 via (2) and separates phase errors (via Φ) from amplitude loss (via Γ).

Implication. Under identical environmental witnesses (T_1 , T_ϕ , link controls), any residual, signed shift of T_2 with height swaps or gradient permutations diagnoses the Φ -sector in (5). This makes “information” a *rhythmic state*: it endures while the system stays in tune with the local tempo.

IV — Decoherence as desynchronization

Standard open-system models frame decoherence as entanglement with an environment. In a *rhythmic* view, it is loss of *phase-lock* to the local tempo. Linearizing (2) gives the instantaneous law

$$\frac{dC}{d\tau} = (i\Phi(x) - \Gamma(x))C, \quad T_2(x) \equiv \Gamma^{-1}(x), \quad (7)$$

which separates phase advance (via Φ) from amplitude loss (via Γ).

Minimal closure (scalar + gradient).

$$\Gamma(x) = \Gamma_{\text{env}}(x) + \xi\Phi(x) + b|\nabla \ln \Phi(x)|\Phi(x), \quad (8)$$

This is a first-order phenomenological model used for identification under controlled conditions; it is not asserted as a fundamental dynamical law. with ξ, b dimensionless and $\Gamma_{\text{env}} \geq 0$ (platform/environment). In the terrestrial weak-field regime, small controlled altitude variations can decorrelate $\Delta\Phi/\Phi$ from $\Delta|\nabla \ln \Phi|$ (Appendix A.1–A.3).

Locked vs. unlocked.

- *Locked (entrained)*: $|\Delta\Phi| \lesssim \Delta\Phi_c$ (or relative), smooth gradients $\Rightarrow \Gamma \simeq \Gamma_{\text{env}}$, T_2 maximal.
- *Unlocked (desync)*: drift/gradient out of band $\Rightarrow$ a signed increase of Γ par (8), an exponential decay of C .

Differential observable (testable prediction). To first order,

$$\frac{\Delta T_2}{T_2} \simeq -\frac{\Delta\Gamma}{\Gamma} \approx -\xi \frac{\Delta\Phi}{\Phi} - b\Delta|\nabla \ln \Phi| - \Delta\left(\frac{\Gamma_{\text{env}}}{\Gamma}\right), \quad (9)$$

where the last term is *vetoed* by witnesses (T_1 , T_ϕ) and link controls. The *scalar* signature is a slope versus $\Delta\Phi/\Phi$ (coefficient ξ); the *gradient* signature is a slope versus $\Delta|\nabla \ln \Phi|$ (coefficient b). Permutation plans and site $\leftrightarrow$ reference swaps (Appendix) disambiguate these contributions.

V — The breathing of information

Information persists by *breathing* with the local tempo: each cycle of $\Phi(x)$ renews phase and keeps probability assignments consistent. The modulus decays with Γ , but the seat of persistence is the phase advance set by Φ .

Coarse-grained tempo and phase. Over a window $\Delta\tau$, define the running average and fluctuation

$$\bar{\Phi}_{\Delta}(\tau) \equiv \frac{1}{\Delta\tau} \int_{\tau}^{\tau+\Delta\tau} \Phi(x(\tau')) d\tau', \quad \delta\Phi_{\Delta}(\tau') \equiv \Phi(x(\tau')) - \bar{\Phi}_{\Delta}(\tau).$$

The phase in (2) splits as

$$\varphi(\tau) = \int_{\tau_0}^{\tau} \Phi d\tau' = \bar{\Phi}_{\Delta}(\tau - \tau_0) + \int_{\tau_0}^{\tau} \delta\Phi_{\Delta}(\tau') d\tau'. \quad (10)$$

Ensemble dephasing from slow breathing. If $\delta\Phi_{\Delta}$ is (locally) stationary with low-frequency power $S_{\Phi}(0)$, Ramsey contrast over time T obeys the Gaussian-noise estimate

$$|C(T)| \approx \exp\left[-\frac{1}{2} S_{\Phi}(0) T\right] \times \exp\left[-\int \Gamma d\tau\right], \quad \Gamma_{\text{breath}} \equiv \frac{1}{2} S_{\Phi}(0), \quad (11)$$

so the *breathing* variance induces an effective dephasing rate Γ_{breath} in addition to the loss channel Γ .

Emergent classicality = loss of entrainment.

- *Quantum regime (entrained):* $\text{rms}[\delta\Phi_{\Delta}]/\bar{\Phi}_{\Delta} \lesssim (\Delta\Phi/\Phi)_c \Rightarrow |C| \simeq 1$ over many cycles; phase averages track $\bar{\Phi}_{\Delta}$.
- *Classical regime (desync):* $\text{rms}[\delta\Phi_{\Delta}]/\bar{\Phi}_{\Delta} \gtrsim (\Delta\Phi/\Phi)_c \Rightarrow \Gamma_{\text{breath}} T \gg 1$ on experimental windows; fringes wash out and only coarse averages survive.

Operational readout.

$$\Gamma_{\text{eff}} \equiv \Gamma + \Gamma_{\text{breath}} \simeq \Gamma_{\text{env}} + \xi \Phi + b |\nabla \ln \Phi| \Phi + \frac{1}{2} S_{\Phi}(0), \quad (12)$$

so that (i) stabilizing environmental witnesses compresses Γ_{env} , (ii) permutations in height/gradient identify the $\{\xi, b\}$ sector, and (iii) residual low-frequency phase noise bounds $S_{\Phi}(0)$ —a direct metrological handle on the breathing of Φ itself.

noindent

textbfTakeaway (operational). Observable “collapse-like” loss of contrast can be modeled as a failure of phase-lock: when tempo fluctuations or gradients exceed capture, entrainment fails and coherence decays.

VI — Towards a temporal thermodynamics of information

Entropy can be read as *phase dispersion* relative to the local tempo. Coherence persists when phase advance tracks $\Phi(x)$; it decays when dispersion grows.

Phase-entropy from tempo fluctuations. Let $\delta\Phi(\tau) \equiv \Phi(x(\tau)) - \langle\Phi\rangle$ on a window T . Define the phase-entropy functional

$$S_{\varphi}(T) \equiv \frac{1}{2} \text{Var}\left[\int_0^T \delta\Phi(\tau) d\tau\right] = \frac{1}{2} \int_0^T \int_0^T \langle \delta\Phi(\tau) \delta\Phi(\tau') \rangle d\tau d\tau'. \quad (13)$$

If $\delta\Phi$ is (locally) stationary with one-sided PSD $S_{\Phi}(\omega)$, then as T grows

$$\frac{dS_{\varphi}}{dT} \xrightarrow{T \rightarrow \infty} \frac{1}{2} S_{\Phi}(0) \equiv \Gamma_{\text{breath}}, \quad |C(T)| \approx e^{-\Gamma_{\text{breath}} T} e^{-\int \Gamma d\tau}, \quad (14)$$

recovering the breathing contribution in (11) and the effective loss in (12).

Thermal analogy (operational).

- *Temperature-like role:* $S_{\Phi}(0)$ acts as a low-frequency “temporal temperature” for phase; cooling the lab reduces Γ_{env} while shielding/referencing reduces $S_{\Phi}(0)$.
- *Free-energy-like target:* maximize *usable coherence* by minimizing $\Gamma_{\text{eff}} = \Gamma + \Gamma_{\text{breath}}$ subject to control costs (power, bandwidth).

Metrological handle (vertical readout). Define the residual fractional shift after applying the standard GR (gravitational+kinematic) model:

$$\left(\frac{\Delta f}{f}\right)_{\text{res}} \equiv \left(\frac{\Delta f}{f}\right)_{\text{meas}} - \left(\frac{\Delta f}{f}\right)_{\text{GR}} \approx \delta \ln \Phi. \quad (15)$$

For a vertical separation Δh , the GR baseline is (to leading order)

$$\left(\frac{\Delta f}{f}\right)_{\text{GR}} \simeq \frac{g}{c^2} \Delta h \approx 1.1 \times 10^{-16} \text{ m}^{-1} \Delta h,$$

plus kinematic terms. Under the null, $\delta \ln \Phi \simeq 0$. The vertical lever Δh is used as a controlled contrast to test whether T_2/Γ shows a reproducible dependence on the inferred Φ -sector once common-mode environment is gated or vetoed (cf. (9)).

Box — Testable predictions (targets).

(1) *Differential clocks & coherence.* Two identical qubits separated by a controlled vertical lever Δh show a signed shift

$$\frac{\Delta T_2}{T_2} \simeq -\xi \delta \ln \Phi,$$

beyond known noise and after GR subtraction. (Scale of the GR baseline: $1.1 \times 10^{-16} \text{ m}^{-1} \Delta h$.)
Target bound: $|\xi| \lesssim 10^{-3} - 10^{-2}$.

(2) *Inter-site Ramsey (common-mode).* Synchronized Ramsey sequences across $10^2 - 10^4 \text{ km}$ reveal sub-Hz common-mode phase noise compatible with regional Φ fluctuations. Use independent links (fiber/GNSS/free space)

VII — Conclusion: the tempo of knowledge

The Chronon Field $\Phi(x)$ reframes time as *tempo*. Coherence is phase-lock to that local tempo; decoherence is desynchronization. The split

$$\dot{\varphi} = \Phi(x), \quad \frac{dC}{d\tau} = (i\Phi - \Gamma)C, \quad T_2 = \Gamma^{-1},$$

makes falsifiable predictions once standard GR effects are subtracted. In practice:

- *Scalar test* (vertical): signed shifts of T_2 scale with $\Delta\Phi/\Phi \simeq g\Delta h/c^2$ (§ VI).
- *Gradient test*: permutations of controlled gradients probe the b -term in Γ (§ IV–VI).
- *Breathing test*: low-frequency plateaus bound $S_\Phi(0)$ via $\Gamma_{\text{breath}} = \frac{1}{2}S_\Phi(0)$ (§ V–VI).

Fail-fast logic is explicit: pass null-tests and recover stable, signed slopes across platforms, or else report upper bounds on $(\xi, b, S_\Phi(0))$ with preregistered analyses (Annex A.1–A.3). Either outcome is informative: detection calibrates the tempo sector; non-detection tightens its admissible window.

Outlook. The companion paper, *The Collapse of Time — When Decoherence Meets Chronon*, develops the quantitative closure for $\Gamma(x)$, specifies target sensitivities for (ξ, b) , and details experimental signatures linking differential T_2 and clock residuals to the local structure and gradients of $\Phi(x)$.

Appendix — Hierarchical model of Γ

A.1 — Base law and variables

Minimal law (phase-loss split).

$$\boxed{\Gamma(x) = \Gamma_{\text{env}}(x) + \xi \Phi(x) + b |\nabla \Phi(x)|} \quad (16)$$

- $\Gamma(x)$: local decoherence (loss) rate $[\text{s}^{-1}]$.
- $\Gamma_{\text{env}}(x)$: platform/environment noise (phonons, $1/f$, stray fields, residual flux, etc.).
- $\Phi(x)$: local tempo (Chronon Field), $[\text{s}^{-1}]$.
- $\nabla \Phi(x)$: spatial gradient of the tempo (Euclidean norm).
- ξ, b : phenomenological couplings (dimensionless under this series' conventions).

Dimensional note (regressor form). For estimation it is convenient to use

$$\Gamma(x) = \Gamma_{\text{env}}(x) + \left[\xi + b |\nabla \ln \Phi(x)| \right] \Phi(x), \quad (17)$$

which keeps b dimensionless and makes the role of the relative slope $|\nabla \ln \Phi|$ explicit.

Observable (Ramsey / echo). Along a worldline $x(\tau)$ parameterized by effective clock time τ ,

$$C(\tau) = \exp\left[-\int_{\tau_0}^{\tau} \Gamma(x(\tau')) d\tau'\right] \Rightarrow \Gamma \simeq -\frac{d}{d\tau} \ln C(\tau), \quad T_2 \equiv \Gamma^{-1}. \quad (18)$$

Hierarchical structure (controlled random effects). Indices: platform p (cQED, ions, NV, Rydberg), site i , run r :

$$\Gamma_{p,i,r}(x) = \underbrace{\Gamma_{\text{env}}^{(p)}}_{\text{platform intercept}} + \underbrace{\delta_i^{(p)}}_{\text{site effect}} + \underbrace{\rho_r^{(p)}}_{\text{run effect}} + [\xi + b |\nabla \ln \Phi(x)|] \Phi(x), \quad (19)$$

with $\sum_i \delta_i^{(p)} = 0$, $\sum_r \rho_r^{(p)} = 0$ (centering) and the constraint $\Gamma_{p,i,r} \geq 0$.

Weak-field anchor (vertical readout). In the stationary terrestrial regime,

$$\Delta \ln \Phi \simeq \frac{\Delta \psi}{c^2} \approx \frac{g}{c^2} \Delta h \approx 1.1 \times 10^{-16} \text{ m}^{-1} \times \Delta h,$$

which is useful to decorrelate the Φ and $|\nabla \ln \Phi|$ terms via *centimeter-meter* altitude micro-variations Δh .

A.2 — Identification strategy by platform

Principle. Isolate the *scalar* contribution $\xi \Phi$ and the *gradient* contribution $b |\nabla \Phi|$ by orthogonally modulating (i) the relative slope $|\nabla \ln \Phi|$ via small altitude changes Δh (weak-field Earth: $\Delta \ln \Phi \simeq g \Delta h / c^2$) and (ii) the environmental stability so that Γ_{env} is held fixed. The observable is the *signed* change of $T_2^{-1} = \Gamma$ between nearly identical conditions, with witnesses T_1 (relaxation) and T_ϕ (pure dephasing).

Differential identification rule.

$$\frac{\Delta \Gamma}{\Gamma} \approx \underbrace{\Delta \left(\frac{\Gamma_{\text{env}}}{\Gamma} \right)}_{\text{veto by witnesses}} + \underbrace{\xi \frac{\Delta \Phi}{\Phi}}_{\text{scalar mode}} + \underbrace{b \Delta |\nabla \ln \Phi|}_{\text{gradient mode}}, \quad \Delta h \text{ small but controlled.}$$

Multi-platform table (regimes, witnesses, targets).

Platform	Typical T_2 regime	Witnesses T_1/T_ϕ	Lever Δh	Target $ \xi $	Target $ b $
cQED (transmons)	20–100 μs (mK)	Echo/CPMG; stable T_1	0–30 m (external ref)	$\lesssim 10^{-3}$	$\lesssim 10^{-2}$
Trapped ions	1–10 s	$T_1 \gg T_2$; DD	1–100 m	$\lesssim 10^{-4}$	$\lesssim 10^{-3}$
NV centers	1–5 ms (echo)	T_ϕ vs ^{13}C	1–300 m	$\lesssim 3 \times 10^{-4}$	$\lesssim 3 \times 10^{-3}$
Rydberg (vapors)	10–100 μs	Known radiative T_1	1–50 m	$\lesssim 10^{-3}$	$\lesssim 2 \times 10^{-2}$

Table 1 Operational windows (order of magnitude) for joint estimation of ξ and b . Targets reflect achievable bounds on $\Delta \Gamma / \Gamma$ (hence $|\xi|$, $|b|$) with $\sim 10^4$ – 10^6 repeats, witnesses, and null-tests.

Per-platform tactics.

- *cQED*: Stabilize T_1 via bias/flux settings; modulate $|\nabla \ln \Phi|$ using a *reference* (clock/oscillator) translated in height and connected through independent links (fiber vs free-space) for link veto; seek $\Delta \Gamma$ correlated with Δh at fixed T and microwave noise.
- *Trapped ions*: Swap the optical table between discrete Δh levels; separate ξ (binning by site) from b (slope vs Δh); use DD (Uhrig/CPMG) to neutralize low-frequency Γ_{env} .
- *NV centers*: Field campaigns (building/hill/basement); witnesses for B -field, temperature, charge; compare T_2 (echo) and T_2^* (Ramsey) to isolate T_ϕ ; blind permutation of locations.
- *Rydberg*: Short sequences with controlled density and blackbody field; alternate lab floors (elevator); check that radiative T_1 is invariant across elevations.

Null-tests and acceptance criteria.

- Invariance of T_1 under Δh (failure if violated: environment not frozen).
- No correlation of $\Delta \Gamma$ with link parameters (link type, power, bandwidth).
- Persistence under permutations (site $\leftrightarrow$ reference) and interleaved-session decorrelation.
- *Signed* signature: $\Delta \Gamma$ proportional to $\Delta |\nabla \ln \Phi|$ (slope b) and to $\Delta \Phi / \Phi$ (slope ξ), with white residuals.

A.3 — Joint regression plan (preregistered) & robustness

Design (preregistration).

- *Primary outcome*: $y \equiv \ln(\Gamma/\Gamma_{\text{ref}})$ by block (log-ratio stabilises variance).
- *Explanatory variables*: $X_1 \equiv \Delta\Phi/\Phi$ (scalar mode), $X_2 \equiv \Delta|\nabla \ln \Phi|$ (gradient mode).
- *Contrasts*: slopes ξ (on X_1) and b (on X_2). Platform intercept α_p .
- *Randomisation*: blind permutations of Δh levels and run order; site $\leftrightarrow$ reference swaps.
- *Stopping rule*: N fixed by target power, not by interim p -values; no peeking analyses.
- *Preprocessing*: outlier censoring by fixed $\text{MAD} \times k$ rule; no imputation; full logging of witnesses (T , B , RF, link).

Joint model (hierarchical GLS). For measurement m (platform p , site i , run r):

$$y_{pirm} = \underbrace{\alpha_p}_{\text{platform}} + \underbrace{u_{pi}}_{\text{site}}, \quad u_{pi} \sim \mathcal{N}(0, \sigma_u^2) + \underbrace{v_{pir}}_{\text{run}}, \quad v_{pir} \sim \mathcal{N}(0, \sigma_v^2) \\ + \xi X_{1,pirm} + b X_{2,pirm} + \varepsilon_{pirm}, \quad \varepsilon_{pirm} \sim \mathcal{N}(0, \sigma_{pirm}^2), \quad (20)$$

with GLS weights σ_{pirm}^{-2} propagated from the uncertainty on Γ (via the contrast $C(\tau)$). Robust variant: Huber–White on ε .

Compact form (matrix).

$$\hat{\beta} = \arg \min_{\beta} (y - Z\alpha - X\beta)^\top W (y - Z\alpha - X\beta), \quad \beta = \begin{bmatrix} \xi \\ b \end{bmatrix},$$

where Z encodes platform fixed effects, $X = [X_1 \ X_2]$, and $W = \text{Cov}^{-1}(y)$ (GLS).

Bayesian option (direct bounds).

$$\xi \sim \mathcal{N}(0, \tau_\xi^2), \quad b \sim \mathcal{N}(0, \tau_b^2), \quad \sigma_u, \sigma_v, \sigma \sim \text{HalfCauchy},$$

reporting posterior medians, 95% CIs, and upper bounds $|\xi|_{95}$, $|b|_{95}$; Bayes factor $\mathcal{B}_{10}$ for $H_0: \xi=b=0$.

Identifiability & orthogonalisation.

- Balance the design so that $\text{corr}(X_1, X_2) \approx 0$ (micro- Δh sets X_2 , stable epoch sets X_1).
- Center/scale X_1, X_2 per block; include nuisance Z_{link} (link type) if needed, then test its coefficient.

Power (order-of-magnitude, a priori).

$$\text{SE}(\hat{\xi}) \simeq \frac{\sigma_y}{\sqrt{N} \text{sd}(X_1)}, \quad \text{SE}(\hat{b}) \simeq \frac{\sigma_y}{\sqrt{N} \text{sd}(X_2)}.$$

Targets: $\text{SE}(\hat{\xi}) \lesssim 10^{-4}$, $\text{SE}(\hat{b}) \lesssim \text{few} \times 10^{-3}$ (cf. Tab. 1).

Useful differential estimators.

$$\hat{\xi}|_{X_2=0} \approx \frac{\Delta y}{\Delta X_1}, \quad \hat{b}|_{X_1=0} \approx \frac{\Delta y}{\Delta X_2}, \quad (21)$$

implemented as *difference-in-differences* on quasi-twin pairs (same runs, alternated Δh).

Robustness battery (pre-specified).

1. *Decoupling* (CPMG/Uhrig ON/OFF): $\hat{\xi}, \hat{b}$ invariant; only residual Γ_{env} moves.
2. *Altitude crossovers* A/B $\leftrightarrow$ B/A: same slopes, swapped intercepts.
3. *Link independence* (fiber vs free-space vs GNSS): stable ξ, b ; any link coefficient $\rightarrow 0$.
4. *Placebo* (shuffled Δh labels): slopes $\rightarrow 0$ within 95% CIs.
5. *Synthetic injections* ($\delta X_1, \delta X_2$ known): unbiased recovery of ξ, b within $\pm 1\sigma$.
6. *Residuals*: white (Ljung–Box), uncorrelated with witnesses (T , B , RF); Durbin–Watson ≈ 2 .
7. *Heteroscedasticity*: HC3 robustness; GLS reweighting confirmed.

Acceptance/failure (fail-fast).

- *Accept (detection)*: non-zero ξ and/or b and cross-platform coherence (fixed/random-effects meta-analysis); white residuals; null-tests passed.
- *Fail (bounds)*: if null-tests fail or cross-platform instability, report only upper bounds $|\xi|_{95}, |b|_{95}$ from the Bayesian or profile-likelihood analysis.

Outputs to report (uniform).

1. Table of $\{\hat{\xi}, \hat{b}\}$ by platform + meta-slope (CI, BF, p).
2. Robustness tests (statistics, p /BF); slope-by-platform figure.
3. Preregistration file (timestamped DOI) + data/analysis *repro pack*.

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